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CDE2301 Value Creation in Innovation

Jovan Tan
Lecturer, Engineering Design and Innovation Centre

jovan@nus.edu.sg

Week 2 (20 August 2026)
Designing a Value Proposition for All Stakeholders I

The author acknowledge contributions from H. Califano, V. Cha, C. L. Cooney, U. Deshpande, and K. K. Nallur in Adaptive Innovation™: An Entrepreneur's Guide to Technology Innovation. WORLD SCIENTIFIC, 2022. doi: 10.1142/13072.

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1 **Understand Value Proposition**

Understand how to craft benefits-driven Value Proposition Design.
Tools: Value Proposition Canvas and Value Proposition Statement.

2 **Understand Stakeholder**

Understand what and who are the stakeholders.
Tools: Stakeholder Analysis.

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Business Plan

An effective business plan needs to answer the following top 10 questions:

Effective Business Plan

- **What is your current situation?**
- **What is your goal?**
- **How will you reach this goal?**
- **What do you offer?**
- **How big is the business you are entering?**
- **Who are the key players?**
- **How will you differentiate yourself?**
- **What will be your marketing plan?**
- **What are the economics of your business?**
- What is the capital requirement to get your business started?

Source: Prof Virginia Cha 3

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Effective Business Plan

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Week 2:
- **Value Proposition**

Source: Prof Virginia Cha 4

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Value Proposition

Value *is fair return for money*

Proposition *is something offered for consideration.*

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Assembling the Building Block

Value Proposition is a key building block for successful business creation

Jobs To Be Done (JTBD)	Technology Step Changes	Current & Future Scenarios of Your Innovation
Understand "Why" would customers want to buy your solution	*Understand the earlier technology advancements and "How" customers accept and adopt it*	*Map current and future change scenarios and identify where your solution can add value*

Elements to Design a Compelling Value Proposition

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Elements to Design a Compelling Value Proposition

Understanding JTBD

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Source: Raw Pixel

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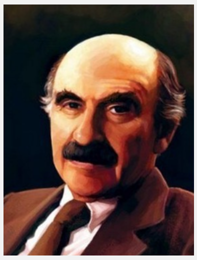
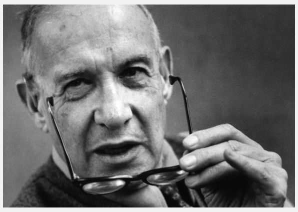
Understanding JTBD

As Theodore Levitt said,

"People Do Not Want A Quarter-inch Drill, They Want A Quarter Inch Hole!"





As Peter Drucker said,

"What the customer buys and considers value is never a product. It is always utility – that is, what a product does for him."

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Understanding JTBD

" Essentially, we hire products to do things for us."

Clayton Christensen, Harvard Business School

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Understanding JTBD

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Source: NUS UCI

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Understanding JTBD

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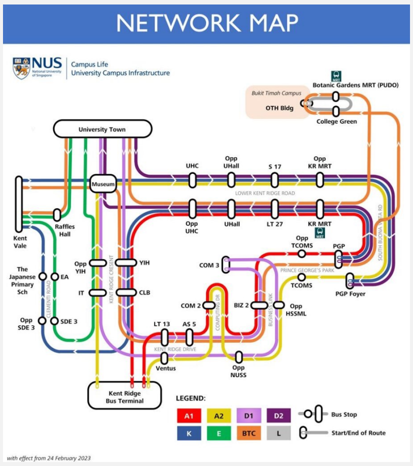
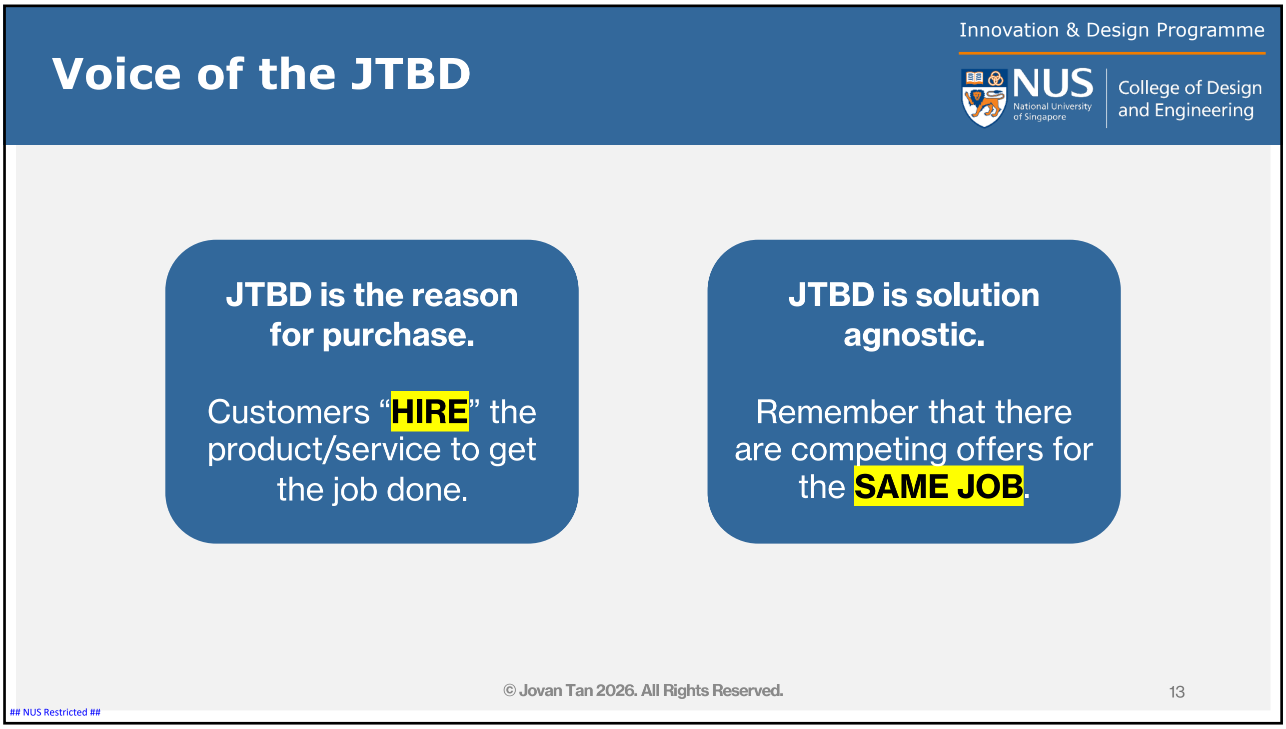


Drive? Cycle? Walk? Skate?

Source: NUS UCI


Voice of the JTBD
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JTBD is the reason for purchase.
Customers "HIRE" the product/service to get the job done.
JTBD is solution agnostic.
Remember that there are competing offers for the SAME JOB.
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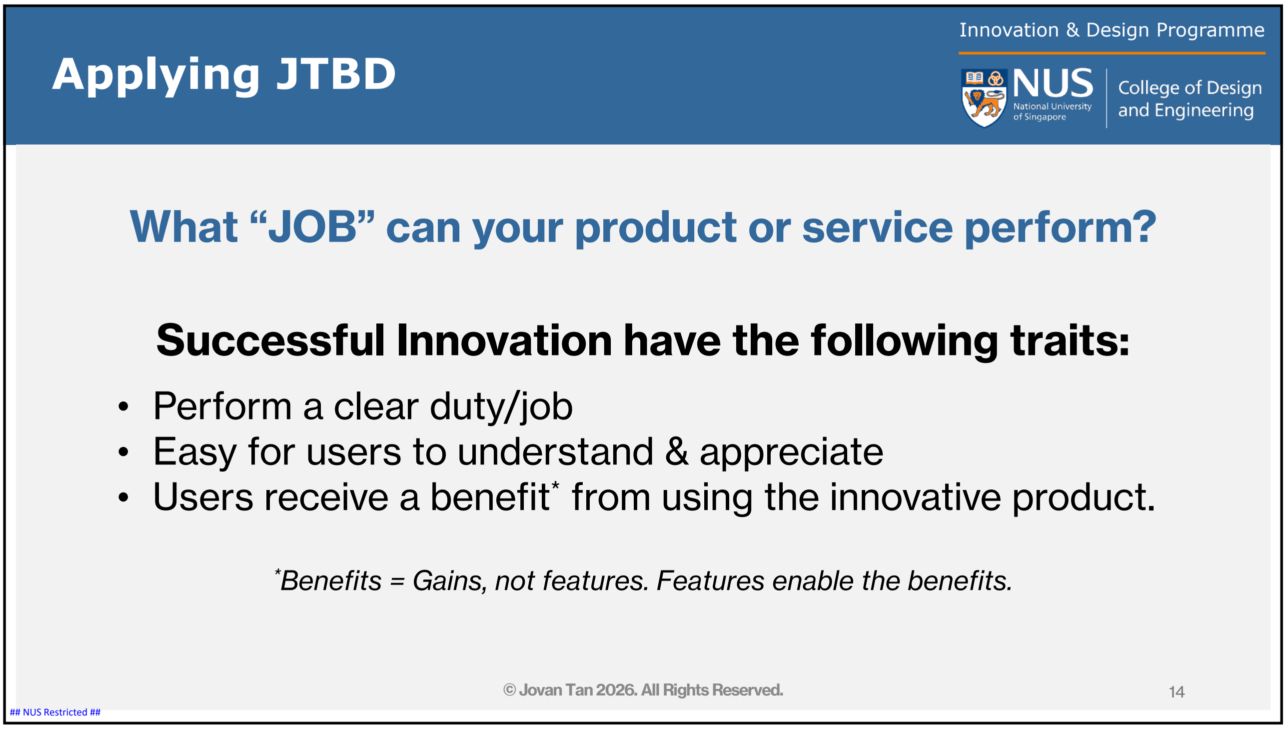
Applying JTBD
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What "JOB" can your product or service perform?
Successful Innovation have the following traits:
• Perform a clear duty/job
• Easy for users to understand & appreciate
• Users receive a benefit* from using the innovative product.
*Benefits = Gains, not features. Features enable the benefits.
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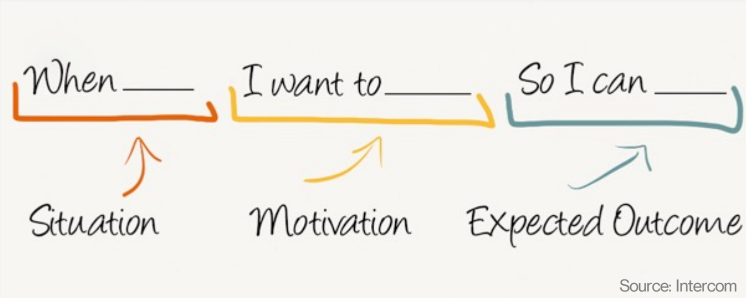
Tool: Using a Job Story



When **<u>I plan for which courses to take in this new semester,</u>**
I want to **<u>be able to visualize my study schedule quickly and accurately</u>**,
So I can **<u>know if I can manage my expected study schedule and add/drop courses accordingly.</u>**

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Assembling the Building Block

Value Proposition is a key building block for successful business creation

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Elements to Design a Compelling Value Proposition

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Purpose

Identify the **winning value proposition** for each technology step change (and, by extension, the product/company that was the iconic winner).

Assessment provides for **potential pattern recognition** or simulation of new ideas.

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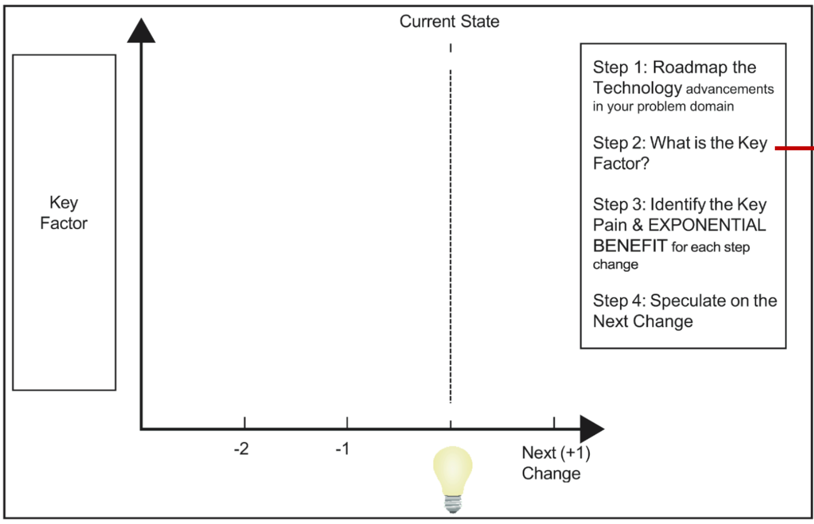
Technological Progress Map



Important Performance Factor

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Case Study: Apple iPod

Source: MacRumours

What opportunity space/industry is the iPod in?

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Opportunity Space: Music Playback

Questions to assess the technology step changes in the music playback space:

1. What are the Step Changes in the technology for music playback domain?
2. What is the important performance factor?
3. For each step change, what undesirable state that could be alleviated from previous state? (possibly an unarticulated problem or latent need)
4. What was the exponential benefit derived?

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Step Changes

Source: The Verge

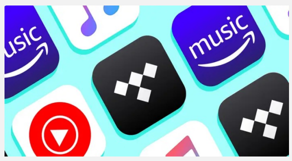
Cassette Players (1970s)

JTBD: Portability
Benefits: Enjoy analog music on-the-go

Source: Apple

MP3 Players (1990s)

JTBD: Access digital music Library on-the-go
Benefits: Enjoy wider selection of pre-loaded music on-the-go



Source: Business Insider

Streaming Services (2010s)

JTBD: Unlimited selections of music on-the-go
Benefits: Choice Freedom

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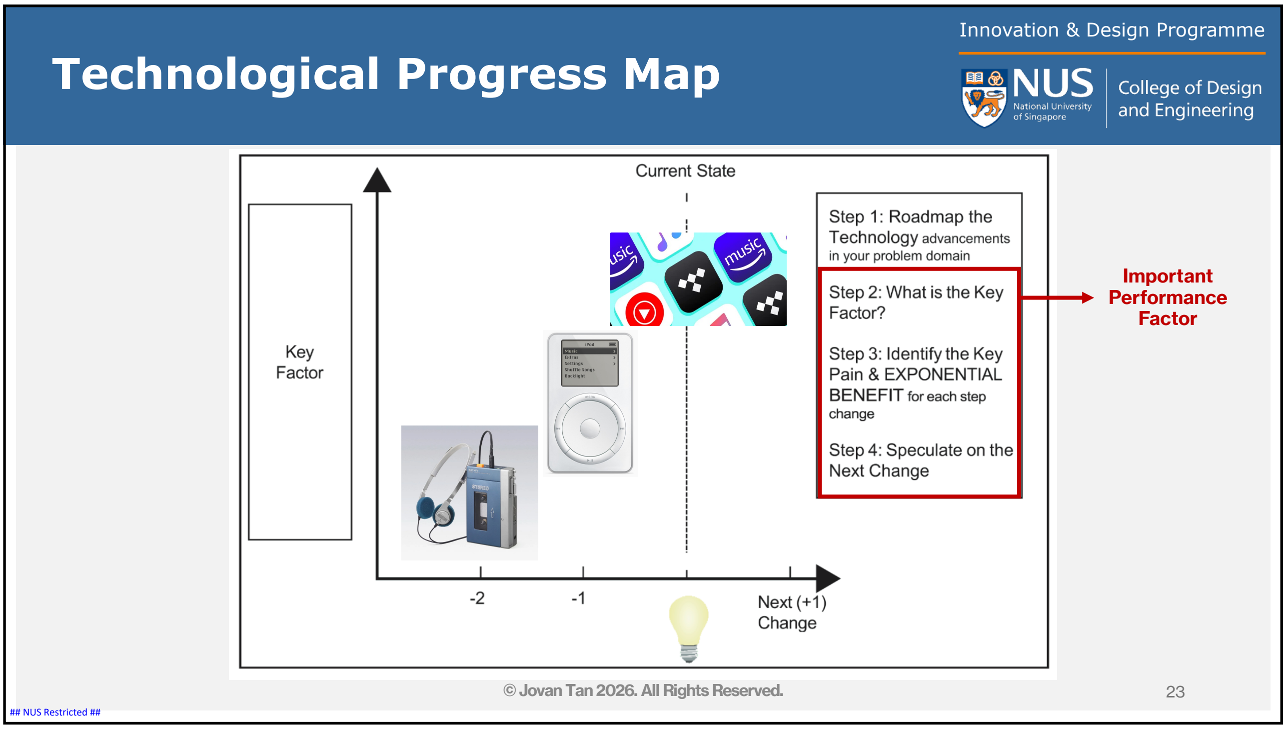
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Technological Progress Map
Current State
Key Factor
Step 1: Roadmap the Technology advancements in your problem domain
Step 2: What is the Key Factor?
Step 3: Identify the Key Pain & EXPONENTIAL BENEFIT for each step change
Step 4: Speculate on the Next Change
Important Performance Factor
-2
-1
Next (+1) Change
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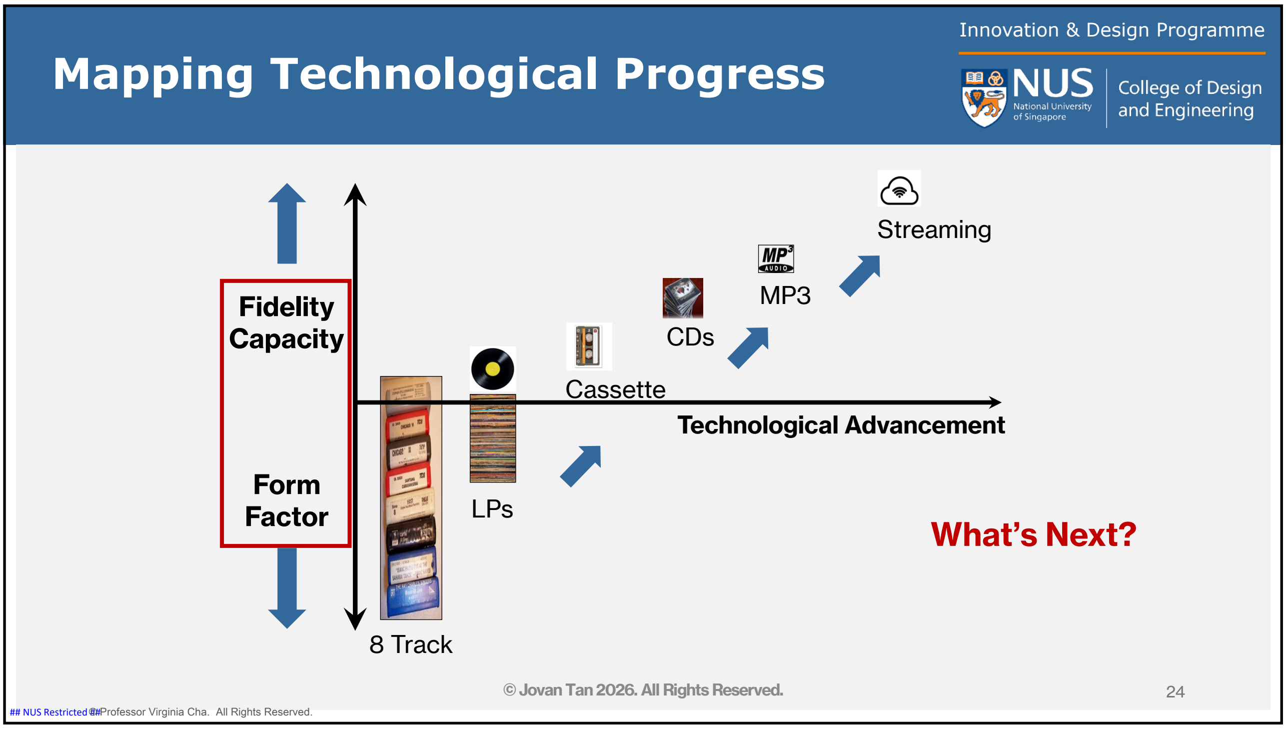
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Mapping Technological Progress
Streaming
MP3
CDs
Cassette
LPs
8 Track
Fidelity Capacity
Form Factor
Technological Advancement
What's Next?
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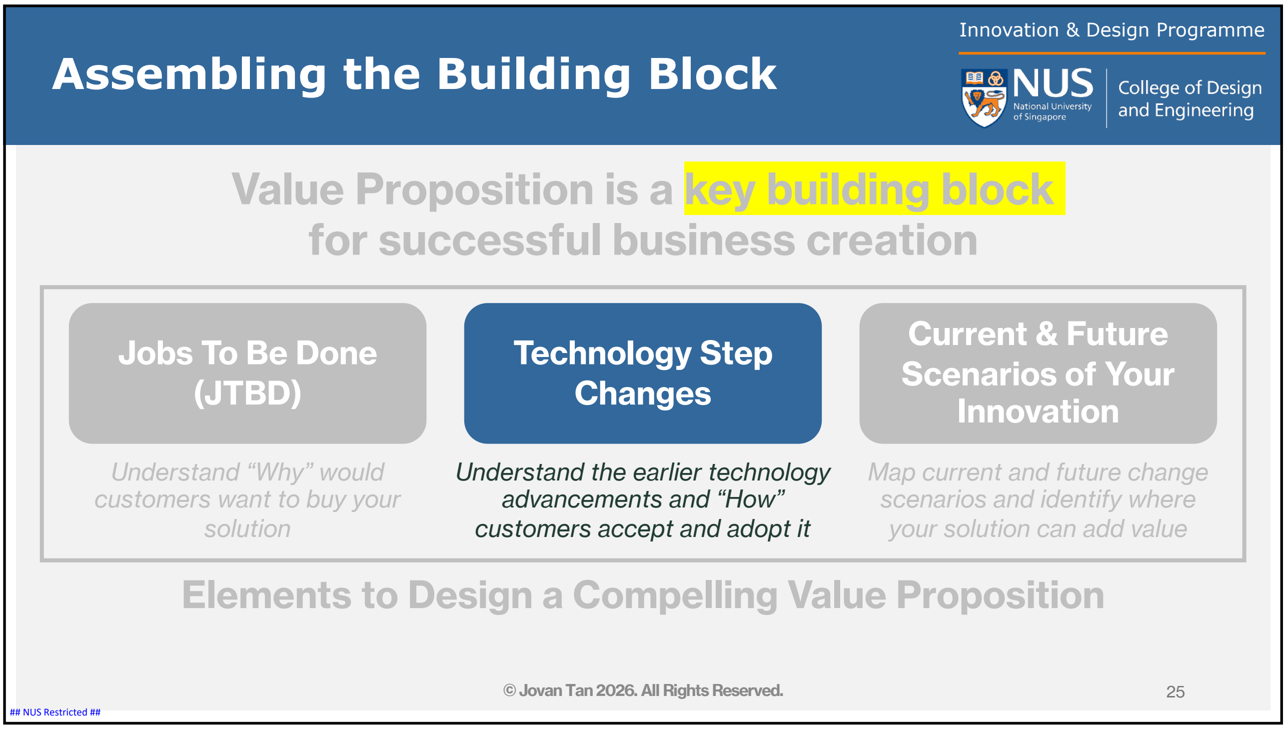
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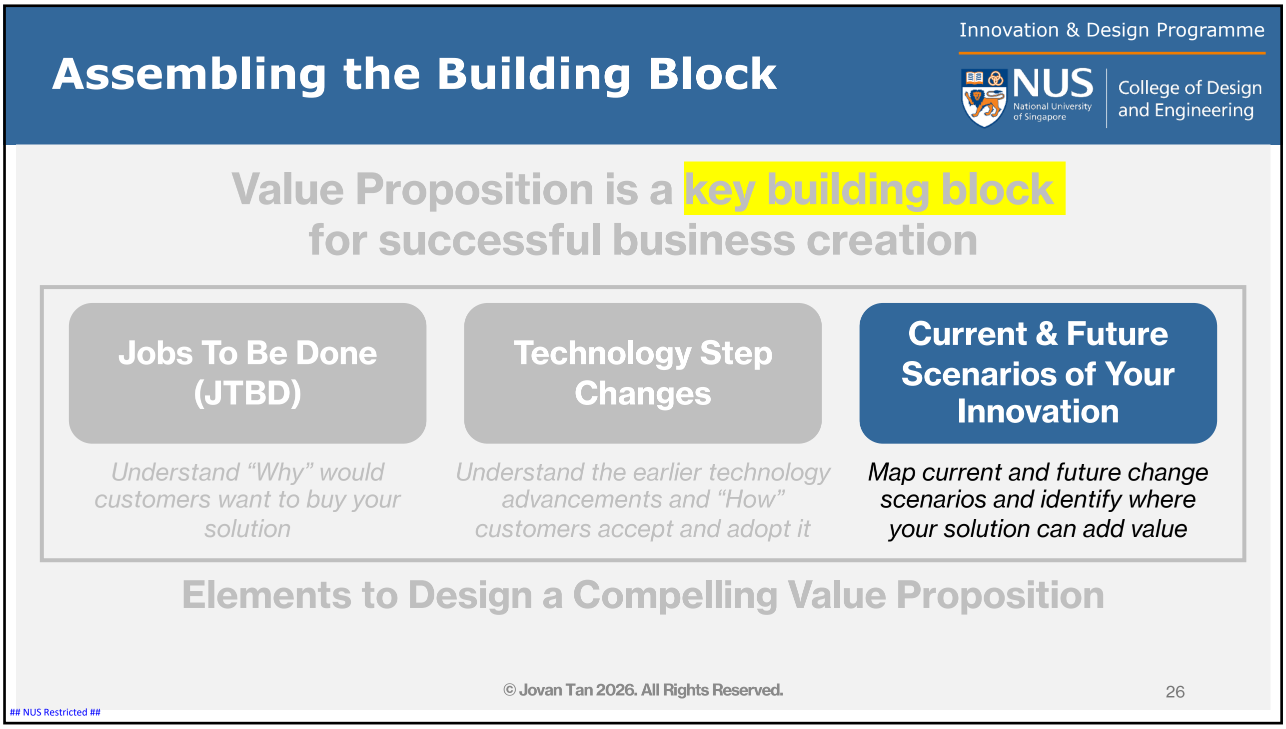
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Purpose

It helps you to:

- understand the relative value you create within the space and **where your innovation can add value**
- identify potential affiliate/associate products/technologies **to complete the solution**.

Thought Process

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1. **In each technology step change, there is a problem worth solving.**
   - We shall term this "problem" as the *undesirable effect.*
   - It could be a latent need.
   - Question: what is the key undesirable effect for your problem space? (aka – what is the underlying problem the customers want solved?)
2. **Do you know what the future undesirable effect looks like?**
3. **Where does your technology fit in the solution space of the future desired effect?**

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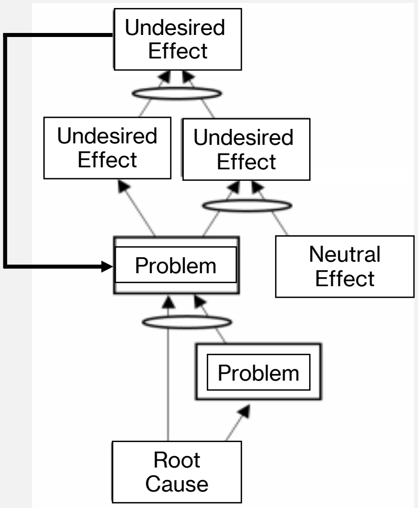
Current Reality Tree



- An undesired effect is observable – typically a symptom the customer wants to be corrected (JTBD).
- The arrows denote causal effects, oval denotes the "and" operator.
- Problem can be observed or hidden.
- Problem can be from the downstream effect of another problem.
- There may be negative feedback loops.
- Root cause may be caused by a desired state from a previous future tree (more on this later).

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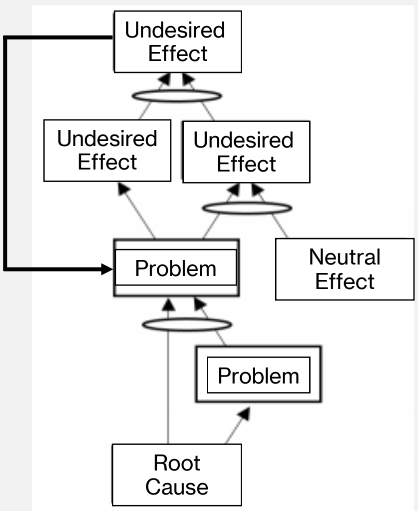
Current Reality Tree



- What Undesired Effect are you targeting?
- Which problem are you solving?
- Does your technology solve this?
- Will solving this problem achieve the desired effect?

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Music Playback Technology Map

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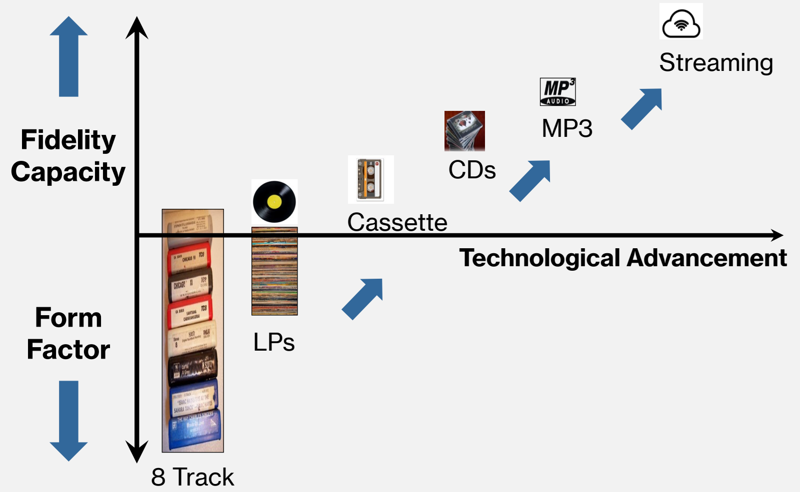

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CRT of MP3

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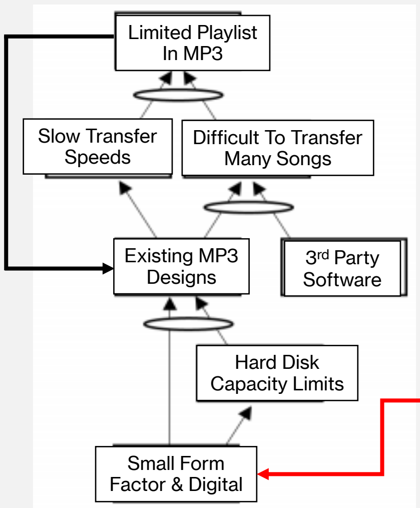

- Small Form Factor was a desired effect from the previous generation, preserved (from Walkman and improved slightly).
- Adopters of MP3 enjoyed the benefits of digital music, but soon felt the constrains of small capacity and difficulties to transfer music.

This was the desired future effect when it was CD players (bulkier & analog).

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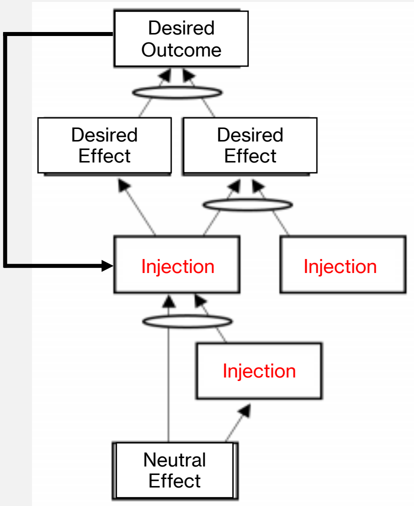
Future Reality Tree



- How can you change the undesirable effects to a Desired Outcome?
- Identify where you can inject a better solution.
- Would the sum of all injections create the desired outcome?
- If not, you may have to add new injections.

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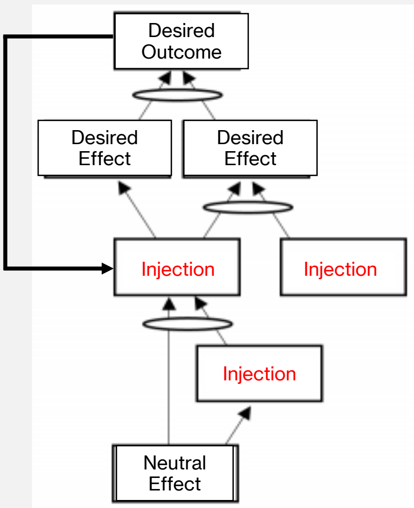
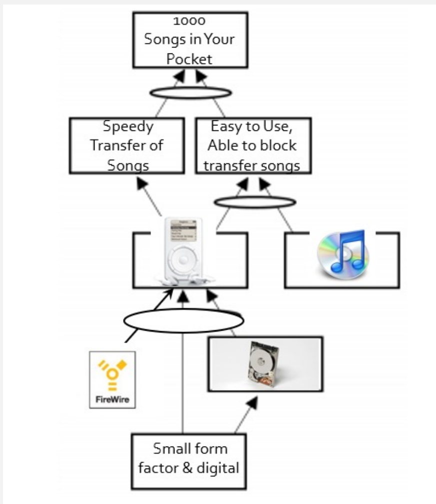
Future Reality Tree





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Summary

1. Technology Step Changes create new opportunities (future desired effects) and new dissatisfactions (current undesired effects).
2. Yesterday's FDEs can become a constraint in current reality and create new undesired effects (or problems) – that's why there is always room for innovation!
3. CRT and FRT are tools from Theory of Constraints to map out the effects and possible interjection points.
4. **The new FDE, combined with a winning business model, becomes the new value proposition offering exponential benefits.**

Note: The trees also contain your assumptions – on the existence of the problem/need and on the desirability of your proposed solution. Thus, we must identify these assumptions and validate them with a rigorous process.

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Elements to Design a Compelling Value Proposition

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Report Back for Week 3

1. **Clearly illustrate and articulate your:**
   1. Jobs To Be Done (JTBD)
   2. Technology Step Changes
   3. Current & Future Scenarios of Your Innovation
2. **Share what your group believe is your opportunities' "VALUE" and "PROPOSITION"**

For CRT/FRT: try to recreate the thought process for the design of your proposed opportunity. By doing so, you will achieve a deeper understanding of the problem space and the opportunity. You can improve on the original CRT/FRT design.